

NEKHA BOSE

Founding Engineer | Full-Stack Engineer | Agentic AI and LLM Systems | Cloud Architecture

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SUMMARY

Founding engineer with 6+ years shipping production agentic AI for **Fortune 100 clients** and high-growth startups. I build **governed AI** where deterministic, versioned logic owns every decision and language models assist only where they are safe, under human-in-the-loop approval, policy-gated actions, and audit-grade traceability. Led architecture and delivery of a four-service AI platform at BreathelT and migrated **Nike** supply-chain tooling to a lakehouse processing 50+ million records a day. **Microsoft Certified Azure AI Engineer**, fluent across React, TypeScript, Python, FastAPI, AWS, and GCP.

CORE COMPETENCIES

Agentic and Multi-Agent AI • Human-in-the-Loop and Policy-Gated Automation • RAG and Retrieval • AI Security and Governance • Full-Stack Engineering • Cloud Architecture • Event-Driven Data Systems • Multi-Tenant Platforms • Technical Leadership

PROFESSIONAL EXPERIENCE

Founding Engineer, BreathelT

Dec 2024 to Present

Remote

- Architected and shipped Ultopia, a production lifestyle-intelligence platform spanning **four deployable services** on a single PostgreSQL source of truth with Redis WebSocket fan-out, taking it from zero to App Store launch in **4 months** and reaching **thousands of users** across client and internal deployments.
- Designed a Claude-powered agent layer, one concierge orchestrating **seven discipline specialists** over a streaming tool-use loop with prompt caching that **cut token cost 60%**, under a hard human-in-the-loop invariant so no action writes without user confirmation.
- Built an event-sourced metric store feeding two scoring engines, a cold-start benchmark scorer and an adaptive personal scorer that weights leading signals over lagging outcomes, cut over from shadow to live across **tens of thousands of user records** with **zero scoring regressions**.
- Built a provider-integration framework spanning **75 providers** across finance, health, wearables, and social, each normalized into canonical metrics through thin config and pure mappers across six auth patterns including SSRF-guarded self-host, with connectors rolling out in phases as providers move from beta to live.
- Hardened the platform with AES-256-GCM encryption at rest, JWT auth with refresh and blacklist, signed OAuth state, per-key rate limiting, and an append-only audit log, with graceful degradation on every external dependency.
- **Led 7 engineers** across architecture, code review, and release on a weekly production cadence, delivering **31 native mobile screens** and a shared PWA on one typed API layer.

Senior Software Engineer, Infosys Limited

Mar 2021 to Dec 2023

On-site, India • Enterprise client accounts

- Delivered production data and platform engineering for **multiple Fortune 100 clients**, including **Nike**.
- Migrated **Nike** legacy supply-chain tooling to an Airflow-orchestrated AWS lakehouse processing **50+ million records a day**, cutting **refresh latency 60%** and **compute cost 30%** by replacing a slow legacy batch cycle with near-real-time refresh every one to two hours.
- Built the ingestion path streaming source events into S3 through transformation, staging, and consumption layers so downstream users always queried clean, near-real-time data.
- Engineered the transformation layer to filter multi-region source data to US-relevant records and segregate freight, shipments, and profiles into separate domains with quality checks before promotion.
- Tuned PostgreSQL query paths and built Tableau dashboards that gave supply-chain stakeholders faster, clearer operational signal for freight and shipment decisions.

Software Engineer, Freelance

Oct 2018 to Mar 2021

Remote

- Delivered end-to-end web and product builds for **5+ clients**, owning architecture, frontend, API integration, and production readiness from scoping through release.

SELECTED PROJECTS

SentinelAI AI-native identity security

github.com/nekhabose/SentinelAI

Treats AI agents as first-class identities, ingesting AWS IAM, CloudTrail, Okta, and Azure AD into a NetworkX identity graph with a **five-factor risk engine** and human-readable explanations. Ships a blast-radius circuit breaker with graduated, reversible auto-remediation, one-click rollback, and agent-specific monitoring across prompt-hash tamper detection, tool allowlisting, and delegation-chain tracking.

Clinara HealthOS *governed clinical-workflow intelligence* github.com/nekhabose/clinara-healthos
Deterministic, versioned logic makes every clinical decision while the LLM only summarizes, drafts, and translates, so critical lab values bypass queues and a red-flag detector can only raise urgency, never lower it. Backed by a transactional outbox, immutable audit, and tenant isolation, with a clinician rule studio to author, simulate, shadow-deploy, and roll back logic without an engineering release.

Helm Finance Agent *agentic SMB finance copilot* github.com/nekhabose/Helm-Finance-Agent
Four production-shape agents, 13-week cash-flow forecasting, AR collections, AP invoice coding, and reconciliation, draft actions and route every effect through a human-in-the-loop approval queue. A deterministic forecast engine stays separate from Claude-generated narrative so numbers remain auditable, over normalized connectors for QuickBooks Online, Plaid, Stripe, and Gusto on a multi-tenant ledger with row-level isolation.

Additional RelayGrid AI, an event-driven freight operations marketplace on a NestJS and Drizzle monorepo, and Call Proof, a bounded inbound voice agent for home-services contractors. Full portfolio at github.com/nekhabose.

EDUCATION

M.S. Information Technology and Management, Illinois Institute of Technology, Chicago, IL Jan 2024 to Dec 2025
GPA 3.90 / 4.0

B.Tech. Information Technology, APJ Abdul Kalam Technological University, Kerala, India

TECHNICAL SKILLS

AI and Agentic Systems Claude and Anthropic SDK, OpenAI, Gemini, Model Context Protocol (MCP), multi-agent orchestration, LangGraph, OpenAI Agents SDK, AutoGen, CrewAI, LangChain, LlamaIndex, tool use and function calling, structured outputs, prompt caching, model routing and fallback, agent memory, bounded autonomy, human-in-the-loop

Retrieval and RAG Hybrid retrieval, reranking, GraphRAG, pgvector, embeddings, semantic search, context engineering

LLM Ops, Eval and Safety Eval-driven development (Braintrust, LangSmith, Ragas), LLM observability and tracing (Langfuse, Arize Phoenix, OpenTelemetry GenAI conventions), guardrails, prompt-injection defense (OWASP LLM Top 10), red-teaming, cost and latency optimization

Languages TypeScript, JavaScript, Python, Go, SQL

Frontend React 19, React Native, Next.js (App Router, Server Components), Expo, Vite, Tailwind CSS, WebSockets and SSE streaming, accessible and responsive design

Backend and APIs Node.js, Express, NestJS, FastAPI, REST, event-driven APIs, async SQLAlchemy, Drizzle ORM, Prisma, Pydantic, Zod, ports and adapters

Data and Streaming PostgreSQL, pgvector, TimescaleDB, MongoDB, Redis, NATS, event sourcing, transactional outbox, Airflow, Tableau, query optimization

Cloud and DevOps AWS (ECS Fargate, RDS, S3, SQS, SNS, KMS, CloudTrail, IAM), GCP, Docker, Kubernetes, Temporal, CI/CD, GitHub Actions, OpenTelemetry, Playwright, Vitest

Security and Compliance Multi-tenant isolation (Postgres RLS), RBAC, SSO (SAML, OIDC), SCIM 2.0, JWT with refresh and blacklist, OAuth 2.0 and OAuth1, AES-256-GCM at rest, BYOK, SSRF and IDOR defense, immutable audit logging, HIPAA-eligible design

CERTIFICATION

Microsoft Certified Azure AI Engineer Associate